

MATH 231 – Linear Algebra I

The Vector Unit, Part 4 ■ The Standard Basis, Span, and Coordinates

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About this document. This is Part 4 of the vector unit. Section numbering runs continuously through all parts, so a reference such as §12.4 points into Part 2 and means exactly what it says:

Part	Sections	Subject
1	§1–§9	vectors, notation, addition, scaling, equality
2	§10–§12	orientation, the dot product, projection
3	§13–§15	Cauchy–Schwarz, metrics, norms
4	§16–	the standard basis, span, coordinates

Assumed from Parts 1–3. Addition and scalar multiplication (§7–§8); the bracket notation $\mathbf{v} = [v_1, v_2]$ and $\mathbb{R}^2$ (§6); the orthogonality test $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ (§11.7); the projection $\text{proj}_{\mathbf{u}}(\mathbf{v})$ and component $\text{comp}_{\mathbf{u}}(\mathbf{v})$ (§12.4); and what it means for two vectors to be parallel (§10.1).

What you should be able to do afterwards. Write down the standard basis vectors and verify that they are orthonormal; express any vector of $\mathbb{R}^2$ as a linear combination of them; say what *span* and *basis* mean; explain what the coordinates of a vector are and why they are unique; and set up the problem of finding coordinates as a linear system.

16. The standard basis of $\mathbb{R}^2$

Since §6 we have written a vector as a bracketed pair, $\mathbf{v} = [v_1, v_2]$, and treated those two numbers as simply given. This section asks where they come from. The answer is that the bracket notation has been making a silent choice all along — and once we see what the choice is, we can start asking what else could have been chosen.

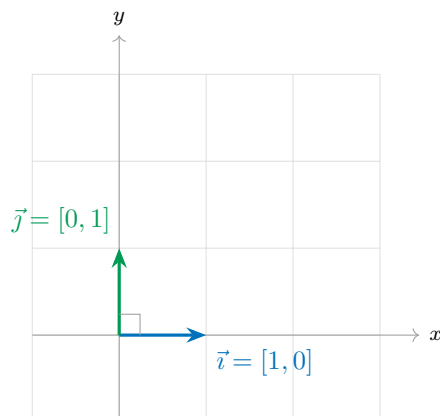
16.1. Two special vectors

Definition 16.1 (The standard basis vectors). The **standard basis vectors** of $\mathbb{R}^2$ are

$$\vec{i} = [1, 0], \quad \vec{j} = [0, 1].$$

They are written with arrows overhead, one of the three conventions of §6.

Geometrically $\vec{i}$ is one step east and $\vec{j}$ is one step north: the two elementary moves, one along each axis.



Three properties, each of which we already have the tools to check.

- **Both have unit length.** $\|\vec{i}\| = \sqrt{1^2 + 0^2} = 1$ and $\|\vec{j}\| = \sqrt{0^2 + 1^2} = 1$, so both lie on the unit circle $\mathbb{S}^1$ of §12.1.
- **They are orthogonal.** Their dot product is $\langle \vec{i}, \vec{j} \rangle = 1 \cdot 0 + 0 \cdot 1 = 0$, so by the orthogonality test of §11.7 they meet at a right angle — as the figure shows, since they lie along the two axes.
- **Neither is a multiple of the other.** No scalar α satisfies $\alpha[1, 0] = [0, 1]$, since the first coordinate would force $\alpha = 0$ and the second would force $\alpha \cdot 0 = 1$. So by §10.1 they are not parallel. Neither one is redundant: you cannot manufacture $\vec{j}$ out of $\vec{i}$ alone.

A collection of vectors that are mutually orthogonal and each of unit length is called **orthonormal** — “ortho” for the right angles, “normal” for the unit lengths. So $\vec{i}$ and $\vec{j}$ form an orthonormal pair, which is as convenient an arrangement as two vectors can have. §16.4 shows what that convenience buys.

A common confusion. This course uses the letters i , j and k for three different things, and you will meet all of them. In the complex numbers lecture, i is the *imaginary unit* with $i^2 = -1$, and i, j, k are the *basis elements* of the quaternions. Here $\vec{i}$ and $\vec{j}$ are *vectors* in $\mathbb{R}^2$. The arrows are what tell them apart, so keep them on.

16.2. Building vectors out of them

Take any two real numbers α and β and form

$$\alpha\vec{i} + \beta\vec{j}.$$

Nothing here is new: it is scalar multiplication from §8 followed by addition from §7. Working it out,

$$\alpha\vec{i} + \beta\vec{j} = \alpha[1, 0] + \beta[0, 1] = [\alpha, 0] + [0, \beta] = [\alpha, \beta],$$

which is a vector in $\mathbb{R}^2$. So every such combination lands inside $\mathbb{R}^2$.

Definition 16.2 (Linear combination). An expression of the form $\alpha\mathbf{u} + \beta\mathbf{v}$, with $\alpha, \beta \in \mathbb{R}$, is called a **linear combination** of $\mathbf{u}$ and $\mathbf{v}$.

The interesting direction is the reverse one. Does every vector of $\mathbb{R}^2$ arise this way, or do these combinations only reach part of the plane? They reach all of it, and the proof is a single line: given any $\mathbf{v} = [v_1, v_2]$, choose $\alpha = v_1$ and $\beta = v_2$, and

$$v_1\vec{i} + v_2\vec{j} = [v_1, v_2] = \mathbf{v}.$$

Nothing in $\mathbb{R}^2$ is missed. Two vectors, scaled and added, generate the entire plane — which is why $\vec{i}$ and $\vec{j}$ deserve to be called the *scaffolding* of $\mathbb{R}^2$: build them once and everything else follows.

16.3. Span and basis

Those two paragraphs proved two inclusions, and together they deserve a name.

Definition 16.3 (Span). The **span** of a collection of vectors is the set of *all* their linear combinations. For $\vec{i}$ and $\vec{j}$,

$$\text{span}\{\vec{i}, \vec{j}\} = \{ \alpha\vec{i} + \beta\vec{j} \mid \alpha, \beta \in \mathbb{R} \}.$$

The first calculation of §16.2 showed $\text{span}\{\vec{i}, \vec{j}\} \subseteq \mathbb{R}^2$, and the second showed $\mathbb{R}^2 \subseteq \text{span}\{\vec{i}, \vec{j}\}$. Both inclusions hold, so

$$\boxed{\text{span}\{\vec{i}, \vec{j}\} = \mathbb{R}^2}$$

and we say that $\vec{i}$ and $\vec{j}$ **span** $\mathbb{R}^2$.

Definition 16.4 (Basis, working definition). A collection of vectors is a **basis** for a space when it spans that space and does so without redundancy — no vector in the collection is a linear combination of the others, so none of them can be dropped. We write

$$\mathcal{B} = \{ \vec{i}, \vec{j} \},$$

and call $\mathcal{B}$ the **standard basis** of $\mathbb{R}^2$.

Remark (Basis and span are different objects). Keep the two apart. The basis $\mathcal{B} = \{ \vec{i}, \vec{j} \}$ is a set containing *two vectors*. Its span, $\text{span}\mathcal{B} = \mathbb{R}^2$, is a set containing *infinitely many* — every vector in the plane. A basis is small; what it generates is large. That is the entire point of a basis: it is a finite description of an infinite set.

Redundancy is the clause worth dwelling on. Spanning alone is easy to achieve by piling on extra vectors: the collection $\{ \vec{i}, \vec{j}, [3, 5] \}$ also spans $\mathbb{R}^2$, but the third vector earns nothing, since $[3, 5] = 3\vec{i} + 5\vec{j}$ was already reachable. A basis is required to carry no such passengers, and §16.1 checked that ours does not: neither $\vec{i}$ nor $\vec{j}$ is a multiple of the other.

Looking ahead. “Without redundancy” is doing real work here and deserves a sharper statement than the one above. It has a proper name — *linear independence* — and a definition that applies to any number of vectors at once, not just two. We will meet it shortly.

16.4. Coordinates

Definition 16.5 (Coordinates). If $\mathbf{v} = \alpha\vec{i} + \beta\vec{j}$, the scalars α and β are the **coordinates** of $\mathbf{v}$ with respect to the basis $\mathcal{B}$.

For the word “the” to be justified, there had better be only one such pair — and there is. Suppose a vector could be written two ways,

$$\alpha\vec{i} + \beta\vec{j} = \alpha'\vec{i} + \beta'\vec{j}.$$

Both sides evaluate componentwise to $[\alpha, \beta]$ and $[\alpha', \beta']$, and two vectors are equal exactly when their coordinates agree (§9). So $\alpha = \alpha'$ and $\beta = \beta'$: the two representations were the same one. Coordinates are **unique**.

Remark (What the bracket notation was hiding). Look again at $\mathbf{v} = [v_1, v_2]$. We have just shown $\mathbf{v} = v_1\vec{i} + v_2\vec{j}$, so the two numbers inside the brackets are exactly the coordinates of $\mathbf{v}$ with respect to $\mathcal{B}$. The notation of §6 was never neutral — it silently reported coordinates in one particular basis. We have been able to ignore this only because $\mathcal{B}$ has been the sole basis in play. Other bases exist, and the same vector has different coordinates in each.

Remark (Why reading them off is free). There is a reason the coordinates turned out to be the components themselves. Because $\mathcal{B}$ is orthonormal, the projection machinery of §12.4 collapses:

$$\text{comp}_{\vec{i}}(\mathbf{v}) = \frac{\langle \vec{i}, \mathbf{v} \rangle}{\|\vec{i}\|} = \frac{1 \cdot v_1 + 0 \cdot v_2}{1} = v_1,$$

and likewise $\text{comp}_{\vec{j}}(\mathbf{v}) = v_2$. So the coordinate of $\mathbf{v}$ along a basis vector *is* its projection component onto that basis vector — the shadow it casts on each axis. Orthonormality is what makes the norms 1 and the cross terms 0, and so makes the arithmetic vanish.

16.5. The reverse problem, and a first linear system

Turn the question around. Instead of being handed α and β and asked what they build, start with a vector $\mathbf{v} \in \mathbb{R}^2$ and treat the scalars as *unknowns*:

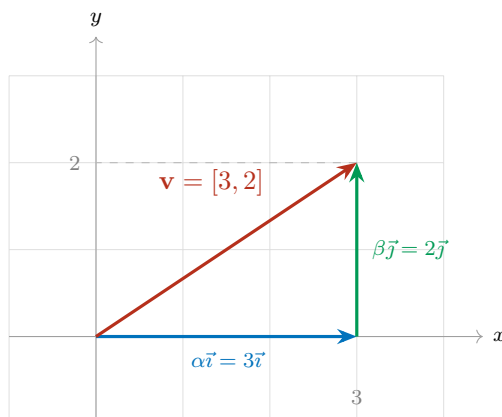
$$\mathbf{v} = \alpha\vec{i} + \beta\vec{j}, \quad \alpha, \beta \text{ to be found.}$$

Two facts already settle that this is a sensible question. Because $\vec{i}$ and $\vec{j}$ span $\mathbb{R}^2$, a solution *exists* — no vector of the plane is out of reach. Because coordinates are unique, the solution is the *only* one. We are not searching hopefully; we are computing something we know is there and know is unambiguous.

The picture

Geometrically the answer is a walk. Starting at the origin, travel α steps along $\vec{i}$ — purely horizontal — and then, from where you land, β steps along $\vec{j}$ — purely vertical. That is the head-to-tail reading of addition from §7.1, and the question is how far to walk in each direction to arrive exactly at the head of $\mathbf{v}$.

Read the other way, the dashed guides drop from the head of $\mathbf{v}$ onto the two axes, and where they land is the answer.



Writing it as a system

Now do it algebraically. Expand the right-hand side componentwise:

$$\alpha\vec{i} + \beta\vec{j} = [\alpha \cdot 1 + \beta \cdot 0, \alpha \cdot 0 + \beta \cdot 1].$$

Setting this equal to $\mathbf{v} = [v_1, v_2]$ and matching coordinates gives two equations in the two unknowns:

$$v_1 = \alpha \cdot 1 + \beta \cdot 0,$$

$$v_2 = \alpha \cdot 0 + \beta \cdot 1.$$

A collection of equations like this — several linear equations sharing the same unknowns, to be satisfied simultaneously — is called a **linear system**. This one barely deserves the name, because the zeros knock out one term in each equation and it collapses on sight:

$$v_1 = \alpha, \quad v_2 = \beta.$$

Example 16.1. For $\mathbf{v} = [3, 2]$ the system gives $\alpha = 3$ and $\beta = 2$, so $\mathbf{v} = 3\vec{i} + 2\vec{j}$ — the walk in the figure above. For $\mathbf{v} = [-2, 5]$ it gives $\mathbf{v} = -2\vec{i} + 5\vec{j}$: two steps west, then five north.

Remark. Do not mistake how easy that was for how easy the question is. The system collapsed because the coefficient of β in the first equation and of α in the second were both 0, and those zeros are precisely the statement $\langle \vec{i}, \vec{j} \rangle = 0$ — the orthogonality we checked in §16.1, reappearing as the arithmetic that makes the work disappear. The standard basis is engineered so that reading coordinates costs nothing.

Looking ahead. Ask the same question against a basis that is *not* the standard one and the zeros are gone. The two equations then genuinely interact, both unknowns appear in both, and the system must actually be solved. That is the general problem of linear systems — how to state it, when it has a solution, how many solutions there can be, and how to compute them — and it will occupy a substantial part of this course. The trivial case above is worth remembering as the one you are always trying to get back to.